1 Problem Description

The values a, b and c can be 0 or 1.

There are 8 triples (permutations) to consider

Find:

$$\sum_{a,b,c} \int_{-1}^{c} (x-a)^b (x+b)^c x^a dx$$

2 Problem Solution

So there are a probably better ways of doing this but im feeling lazy so i will just be doing the simple method of brute force.

When a,b,c = 0,0,0

$$\int_{-1}^{0} (x-0)^0 (x+0)^0 x^0 dx$$
$$= \int_{-1}^{0} 1 dx$$
$$= [x]_{-1}^{0}$$
$$=1$$

When a,b,c = 0,0,1

$$\int_{-1}^{1}(x-0)^0(x+0)^1x^0dx$$
$$=\int_{-1}^{1}xdx$$
$$=\left[\frac{x^2}{2}\right]_{-1}^{1}$$
$$=\frac{1}{2}-\frac{1}{2}$$
$$=0$$

When a,b,c = 0,1,0

$$\int_{-1}^{0}(x-0)^1(x+1)^0x^0dx$$
$$=\int_{-1}^{0}xdx$$
$$=\left[\frac{x^2}{2}\right]_{-1}^{0}$$
$$=-\frac{1}{2}$$

When a,b,c = 0,1,1

$$\begin{aligned}
&\int_{-1}^{1}(x-0)^1(x+1)^1x^0dx \\
=&\int_{-1}^{1}(x^2+x)dx \\
=&\left[\frac{x^3}{3}+\frac{x^2}{2}\right]_{-1}^{1} \\
=&\frac{1}{3}+\frac{1}{2}-\left(\frac{-1}{3}+\frac{1}{2}\right) \\
=&\frac{2}{3}
\end{aligned}$$

When a,b,c = 1,0,0 (repeats a,b,c=0,1,0)

$$\begin{aligned}
&\int_{-1}^{0}(x-1)^0(x+0)^0x^1dx \\
=&\int_{-1}^{0}xdx \\
=&\left[\frac{x^2}{2}\right]_{-1}^{0} \\
=&-\frac{1}{2}
\end{aligned}$$

When a,b,c = 1,0,1

$$\int_{-1}^{1}(x-1)^0(x+0)^1x^1dx$$
$$=\int_{-1}^{1}x^2dx$$
$$=\left[\frac{x^3}{3}\right]_{-1}^{1}$$
$$=\frac{2}{3}$$

When a,b,c = 1,1,0

$$\int_{-1}^{0}(x-1)^1(x+1)^0x^1dx$$
$$=\int_{-1}^{0}(x^2-x)dx$$
$$=\left[\frac{x^3}{3}-\frac{x^2}{2}\right]_{-1}^{0}$$
$$=\frac{5}{6}$$

When a,b,c = 1,1,1

$$\int_{-1}^{1}(x-1)^1(x+1)^1x^1dx$$
$$=\int_{-1}^{1}(x^3-x)dx$$
$$=\left[\frac{x^3}{3}-\frac{x^2}{2}\right]_{-1}^{0}$$
$$=\frac{5}{6}$$

$$\therefore \sum_{a,b,c}\int_{-1}^{c}(x-a)^b(x+b)^cx^adx$$
$$=1+0-\frac{1}{2}+\frac{2}{3}-\frac{1}{2}+\frac{2}{3}+\frac{5}{6}$$
$$=\frac{13}{6}$$